Kavennesh Balachandar

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SUMMARY

Offensive Security Engineer with hands-on experience in application security, penetration testing, vulnerability assessment, and network defense. Experienced in remediating OWASP Top 10 vulnerabilities, strengthening authentication and API authorization, and developing automated security assessment tools. Conducts vulnerability research across HackerOne and Bugcrowd, with expertise in Python, AST taint analysis, MCP server security, and enterprise firewall configuration.

EXPERIENCE

Cybersecurity Analyst (Intern) - Application Security, Community Dreams Foundation | Remote *Aug 2026 – Present*

- Remediated OWASP Top 10 flaws on a production HR platform, tracing injection, broken access control, and authentication weaknesses through validated fixes.
- Executed pre-release vulnerability testing across application surfaces, documenting findings and verifying remediation before release to reduce the risk of exploitable production defects.
- Implemented authentication, role-based access control, and endpoint authorization for a production HR platform, strengthening input validation, error handling, and secrets management.

Graduate Teaching Assistant, Florida International University *Aug 2025 – Apr 2026*

- Audited SQL Server permissions and stored-procedure access paths during database administration labs, teaching secure privilege configuration and query-plan analysis to 60+ students.
- Rebuilt database lab setup and troubleshooting workflows, improving student performance by 20% through clearer instructions, repeatable exercises, and targeted one-on-one support.
- Investigated SQL injection, privilege escalation, and audit-log forensics in database security labs, producing reference guides used by more than 60 students.

Graduate Teaching Assistant, Florida International University *Jan 2025 – Aug 2025*

- Designed attack simulations for packet injection, SSH misconfiguration, and port scanning, improving lab completion by 22% through clearer instructions and pre-lab primers.
- Configured and troubleshot iptables, firewalld, and Snort in Linux security labs, creating MITRE ATT&CK mapping guides that earned 95% positive feedback.
- Automated lab submission processing with custom scripts and supported Wireshark traffic analysis and Snort/Suricata rule-writing exercises, reducing grading turnaround time by 40%.

Network Security Analyst, Inter Deccan Solutions | Chennai, India *Apr 2022 – Mar 2024*

- Analyzed production traffic with Nmap, Wireshark, and tcpdump to identify network misconfigurations, documenting vulnerability-assessment findings in monthly risk reports for remediation.
- Removed 25+ redundant firewall rules and standardized TLS 1.2/1.3, eliminating rule shadowing and reducing false-positive alert volume by 20% overall.
- Configured Cisco Secure Firewall 3100 zone policies, NAT, and ACLs across three enterprise sites, reducing unauthorized access by 40% within six weeks.

PROJECTS

VPN Blocker: Detection & Prevention of VPN Tunneling - Peer-Reviewed *2024*

- Evaluated VPN-tunnel detection using flow metadata, handshake timing, and packet-size entropy without decryption, achieving 90–95% accuracy and reducing policy violations 50–60%.

MCPScan - MCP Server Security Scanner *Jul 2026 – Aug 2026*

- Built nine MCPScan rules combining AST taint analysis and Docker-sandboxed probing, testing 77 servers and 1,289 tool definitions with 822 automated tests.

Cyber Range Lab - Penetration Testing Environment *Apr 2026 – May 2026*

- Built a Docker-isolated cyber range with DVWA and OWASP Juice Shop behind UFW segmentation, validating OWASP Top 10 exploitation at 95%+ success.
- Automated cyber-range provisioning with Bash infrastructure scripts, reducing environment setup from 30 minutes to five and enabling repeatable, isolated penetration-testing exercises.

PentestFlow - Automated Assessment Pipeline *Jan 2026 – Feb 2026*

- Developed PentestFlow to parse Nmap XML results and correlate discovered services with CVEs across 50+ hosts, reducing manual assessment effort by 40%.

- Automated reconnaissance, host discovery, port scanning, and service enumeration in PentestFlow, standardizing repeatable assessment phases and cutting overall assessment time by 30%.

CERTIFICATIONS AND AWARDS

Certifications: OSCP (Current), CompTIA Security+ (SY0-701), Certified Red Team Analyst (CRTA), eJPT, INE Certified Cloud Associate (ICCA), CCSP-AWS (Merit), CNSP (Merit)

Awards & CTF: Amazon AppSec CTF - NAMER Region, Eraksha Cybersecurity Competition - 1st Place National, Kavach Hackathon - National Qualifier

EDUCATION

Florida International University - Master of Science, Cyber Security *Aug 2024 – Apr 2026*

Anna University - Bachelor of Engineering, Computer Science *Nov 2020 – Mar 2024*

SKILLS

Offensive Security & Penetration Testing: Penetration Testing, Vulnerability Assessment, OWASP Top 10, SQL Injection, Cross-Site Scripting (XSS), Command Injection, Privilege Escalation, Reconnaissance, Port Scanning, Exploit Validation

Application & API Security: Source Code Review, Authentication & Authorization Testing, Broken Access Control, IDOR, OAuth Security, API Security Testing, Role-Based Access Control (RBAC), Input Validation, Secrets Management

Security Research & Automation: AST Taint Analysis, MCP Server Security, Vulnerability Research, CVE Correlation, Security Tool Development, Automated Security Testing, Python, Bash

Network & Infrastructure Security: Nmap, Wireshark, tcpdump, Cisco Secure Firewall, Access Control Lists (ACLs), Network Address Translation (NAT), TLS 1.2/1.3, VPN Tunnel Detection, iptables, firewalld

Security Monitoring & Threat Analysis: Snort, Suricata, IDS Rule Writing, Deep Packet Inspection, MITRE ATT&CK, Network Traffic Analysis, Audit Log Forensics

Security Testing Environments & Tools: Docker, DVWA, OWASP Juice Shop, Linux, SQLMap, Hydra, PostgreSQL, Supabase